

|   |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                     |
|---|-----|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 2 | (a) | (i)  | <p><math>N</math>: the number of molecules in the ideal gas</p> <p><math>m</math>: the mass of an ideal gas molecule.</p> <p><math>V</math>: the volume of the ideal gas or volume occupied by the ideal gas (molecules)</p> <p>Note: volume of the container (holding the gas) is accepted, but discouraged, since the question does not mention any container for the ideal gas.</p> <p><math>\langle c^2 \rangle</math>: mean square speed of the ideal gas molecules.</p> <p>“Average of the squared velocities” is accepted</p> <p>(-1 for each mistake, max -2 marks.)</p> <p>molecules/ particles/ atoms are all accepted but make sure it is used consistently throughout.</p>                                                                                                                                                                                                                                                          | B2                  |
|   |     | (ii) | <p><math>p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle \Rightarrow pV = \frac{1}{3} Nm \langle c^2 \rangle</math> ----- (1)</p> <p>Substitute <math>pV = NkT</math> into (1): <math>\frac{1}{3} Nm \langle c^2 \rangle = NkT</math></p> <p><math>\frac{3}{2} \times \frac{1}{3} m \langle c^2 \rangle = \frac{3}{2} kT</math></p> <p>mean kinetic energy of a molecule <math>\langle E_k \rangle = \frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT</math></p> <p>Internal energy of the ideal gas, <math>U</math> = total KE of ideal gas <math>= N \langle E_k \rangle = \frac{3}{2} NkT</math></p> <p>OR:</p> <p><math>p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle</math> ----- (1)</p> <p><math>pV = NkT</math> ----- (2)</p> <p>Internal energy of the ideal gas,</p> <p><math>U</math> = total KE of ideal gas <math>= N \langle E_k \rangle = \frac{1}{2} Nm \langle c^2 \rangle = \frac{3}{2} pV = \frac{3}{2} NkT</math></p> | <p>B1</p> <p>B1</p> |
|   | (b) | (i)  | <p>From Formula list,</p> <p><math>\langle E_k \rangle = \frac{3}{2} kT</math></p> <p><math>\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT</math></p> <p><math>T = \frac{m \langle c^2 \rangle}{3k}</math></p> <p><math>= \frac{40u \langle c^2 \rangle}{3k}</math></p> <p><math>T = \frac{(40 \times 1.66 \times 10^{-27})(2380)^2}{3(1.38 \times 10^{-23})}</math></p> <p><math>T = 9080 \text{ K}</math></p> <p>OR:</p> <p><math>\langle E_k \rangle = \frac{3}{2} kT</math></p> <p><math>\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT</math></p> <p><math>\frac{1}{2} m_R \langle c^2 \rangle = \frac{3}{2} RT</math>, where <math>m_R</math> = molar mass</p> <p><math>T = \frac{m_R \langle c^2 \rangle}{3R}</math></p> <p><math>= \frac{0.040 \langle c^2 \rangle}{3R} = \frac{(0.040)(2380)^2}{3(8.31)}</math></p> <p><math>= 9090 \text{ K}</math></p>                                                                       | <p>B1</p> <p>B1</p> |

|  |  |      |                                                                                                                                                                                                                                                                                                                                                                                                                         |                     |
|--|--|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
|  |  |      | (mass of the argon atom is approximately the same as mass of the argon nucleus, it does not matter whether you take into account the mass of electrons)                                                                                                                                                                                                                                                                 |                     |
|  |  | (ii) | <p>Argon-40 is highly likely that the argon formed from the nuclear decays will be present in the Moon's atmosphere,</p> <p>since the <b><u>temperature on the surface of the Moon is much lower</u></b> than temperature calculated in (b)(i),</p> <p>the majority of argon-40 atoms would have speeds lower than the escape speed from Moon, <b><u>and are unable to escape from the surface of the Moon.</u></b></p> | <p>B1</p> <p>B1</p> |
|  |  |      | Total                                                                                                                                                                                                                                                                                                                                                                                                                   | 8                   |

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(a)

